
Listing 3: Running the benchmark suite, which writes a run directory holding the per-case results and a manifest. Both caveats are load-bearing for anyone recomputing the reported figures.

```
# The release profile is not optional for a timing run: a debug build of  
# the extension is one to two orders of magnitude slower, and measures the  
# build profile rather than the solver.
```

```
uv sync --extra benchmark && uv run maturin develop --release
```

```
# --reps and --mc set the depth. The suite phase runs HALF the requested  
# repetitions, so 2, 4 and 10 time each case 1, 2 and 5 times; both counts  
# are recorded per rung as reps_requested and reps_effective, and the  
# figures in Validation are indexed by the effective value -- to reproduce  
# the reported depth-50 rung, pass --reps 100.
```

```
uv run python -m oracles.cli run --reps 2
```

```
# Four-axis regression gate against the pinned baseline.
```

```
uv run python -m oracles.cli gate
```
